08 Ch 08 – Mutual Funds

Asked in last 3 papers: 7 items, all numerical – NAV and NAV per unit (Sep 25 MCQ, Jan 26 Q6a), holding-period return payout vs reinvestment (Sep 25 Q3b, Q5b with loads, tax, inflation), dividend vs bonus plans (Jan 26 Q1a, May 26 Q3b), closed-ended fund at premium/discount (May 26 Q4b).

1, 1.1–1.5 Introduction, mutual benefits, what is a MF, who can invest, how it works, stocks or MF

- → *MF = a **trust** that pools savings of investors with a common financial goal, through issue of **units**; investor becomes **part owner** of the fund's assets.*
- → *Managed by **professional money managers** as per the scheme's stated investment objective: shares, debentures, bonds and money-market instruments (CP, CD, T-Bills).*
- → *Net income + capital appreciation shared among unit holders **in proportion to units** held. Fund house charges fees based on the **value of assets**.*
- → *Anybody with an investible surplus of a **few thousand rupees** can invest.*
- → ***Stocks or MF?** Diversification needs 15–20 stocks (₹ 5,000 can't be spread); brokerage eats returns on small lots; SIP (e.g. ₹ 1,000 a month) gives discipline; stock picking needs time, inclination and skill; churning costs capital gains tax (MFs do not pay capital gains tax when they sell a stock). Risk-averse investors → MF.*

Establishment / structure of a MF (TYK T1 — ICAI answer):

- → ***SEBI registration** before collecting funds; all MFs under the same regulations, SEBI monitoring and inspection.*
- → ***Sponsor** (like a promoter: gives initial capital, appoints trustees) — body corporate in financial services for **not less than 5 years**, financially sound, fit party.*
- → ***Trust** (or trustee company) under the Indian Trusts Act; trust deed executed by sponsor in favour of trustees, registered under the **Indian Registration Act, 1908**, with Third Schedule clauses. Trustee holds MF property in trust for unit holders. At least **two-thirds** of trustee directors independent of sponsor.*
- → ***AMC** — SEBI-approved, limited liability company, **minimum net worth ₹ 100 crore**; sponsor contributes **at least 40%** of AMC net worth; **50%** of AMC directors independent. AMC can be terminated by **majority of trustees** or **75% of unit holders** of the scheme. Charges investment management and advisory fees within ceiling + gets operating expenses reimbursed.*
- → ***Custodian** — SEBI-registered, independent of sponsor and AMC, holds securities (Depository Participant for demat).*

2, 2.1 Classification; Functional classification

Asked: May 2026 Q4(b) 4m (closed-ended)

*Three classifications: **Functional**, **Portfolio**, **Ownership**.*

Point	Open-ended	Close-ended	Interval
Entry/exit	Any time	Invest at NFO or buy on stock exchange after listing; exit by selling on exchange, at expiry, or in repurchase period	Purchase and redemption only in pre-specified intervals (monthly, quarterly, annually) at NAV-based prices
Capital / life	Unlimited capital, indefinite redemption period	Limited life; corpus liquidated at end	Maturity not defined
Listing	—	Listed	Not required (in-built redemption window)
Fresh units	Continuously	Only at NFO	During interval period at NAV-based prices

→ *Interval = combination of open and close-ended; differs from close-ended on:* ***not listed***, ***fresh issue*** *in interval,* ***maturity not defined***.

→ *A close-ended fund trades through the day at prices that may be* ***more or less than its NAV*** *(SM 3.9).*

Note (app-written): for May 2026 Q4(b) type — market price of a close-ended unit = NAV × (1 + premium %) or NAV × (1 − discount %); investor's return is on market price paid/received, the fund's growth is on NAV.

2.2, 2.2.1, 2.2.2 Portfolio classification — Equity and Debt funds

Equity, Debt and Special funds. Equity holder profits via price rise or dividends and bonus; can lose the whole value but is not liable for company debts.

Fund	Invests in / aim	Suits
Growth	*Long-term capital appreciation*	*Long-term investors*
Aggressive	*Super-normal returns: start-ups, IPOs, speculative shares*	*Risk takers*
Income	*Safe high-dividend stocks, high-yield money market instruments*	*Current income seekers*
Bond *(debt)*	*G-bonds, corporate/convertible debentures, money market; less volatile than equity, regular income; used to diversify*	*Tax-free/steady income*
Gilt *(debt)*	*T-Bills and dated securities of State and Central Govts*	—

Risks of bond funds:

→ ***Interest rate risk*** *— bond value moves inversely to interest rate (YTM).*

→ ***Credit risk*** *— issuer defaults on interest/principal; higher for lower credit rating.*

→ ***Prepayment risk*** *— issuer calls callable bonds early when rates fall and refinances at lower rate.*

→ ***Reinvestment risk*** *— maturity proceeds reinvested at lower rate in falling-rate scenario (favourable when rates rise).*

2.2.3 Special funds

→ ***Index*** *— invest in index stocks in the same weightage; passive, low cost; deliver approx. the index return (e.g. NIFTY 50 Index Fund).*

→ ***International*** *— raises money in India, invests globally.* ***Offshore*** *— located outside India, raises money globally to invest in India.*

→ ***Sector*** *— entire fund in one sector (infrastructure, IT, pharma).*

→ ***Money market*** *— debt-oriented; preservation of capital, liquidity, moderate income; CP, CD, T-Bills, G-Secs; park surplus for short periods; less interest-rate sensitive, lower volatility than long-dated bond funds.*

→ ***Fund of Funds*** *— invest in units of other MF schemes (same or other fund house), e.g. Indian MFs investing in international MF schemes.*

→ ***Capital protection oriented*** *— protection comes from* ***portfolio structure***, *not bank guarantee/insurance; SEBI:* ***close-ended, listed***, *portfolio structure* ***mandatorily rated***; *major part in highly rated debt, rest in equity. Not the same as "capital guaranteed".*

→ ***Gold*** *— track gold; generally ETFs, listed; bullion exposure without physical delivery.*

→ ***Quant*** *— data-driven, rule/model-based selection and timing via automated programme; manager checks robustness and monitors models. Differs from index fund manager (who fully hands off to the index).* ***Pros:*** *removes human bias, consistent strategy, lower expense ratio (passive).* ***Con:*** *models tested on historical data — past trends can't be used blindly.*

Note (app-written): SM text says "Special Funds are of four types" but lists nine, (a) to (i) (SM p. 8.6–8.9). List all nine.

2.3–2.4 Ownership classification; Direct plans

- → **Public sector** MF — sponsored by a public sector company/bank. **Private sector** — by a private company. **Foreign** — by a foreign company operating from India, raising and investing funds in India.
- → **Direct plans** (SEBI, **1 January 2013**): AMC charges **no distribution expenses, trail fees, transaction charges** → NAV generally **higher** than regular plan; have performed better to the extent of lower fees. Before this, direct investments went into the regular plan with a single NAV.

3, 3.1–3.2 Types of schemes: Balanced; Diversified equity; income options

- → **Balanced** — strategic allocation to debt (stability) and equity (growth); limited equity exposure; moderate returns.
- → **Diversified equity** — wide array of stocks, low concentration risk:
  - → **Flexicap** — any proportion across market caps, no restriction; true diversification, flexibility.
  - → **Multicap** — at least **25% each** in large, mid and small cap; remaining 25% at manager's discretion.
  - → **Contra** — out-of-favour, under-valued companies; blends growth and value; aggressive risk appetite.
  - → **Index** — same securities and proportion as benchmark (Sensex, Nifty); earns market return.
  - → **Dividend yield** — high dividend yield (DPS ÷ market price) stocks, above index yield; less volatile than growth stocks; medium risk; not always resilient in short-term corrections.

Option	What the investor gets
Growth / cumulative	No intermittent income; return accumulates, paid on withdrawal as **capital gain**.
Dividend payout	Dividend paid out; **NAV falls** to the extent of dividend and statutory levies.
Dividend re-investment	Dividend reinvested at **ex-dividend NAV** → additional units.

3.3–3.8 ELSS, Sector, Thematic, Arbitrage, Hedge, Cash funds

- → **ELSS** — tax saving under **Sec 80C**; **lock-in 3 years**; better return potential than traditional tax-saving instruments; via SIP avoids year-end lump sum and gives averaging.
- → **Sector** — ride industry cycles; good returns if timing is right; more vulnerable on downside than diversified funds; only a limited portion of portfolio; not for first-time investors. E.g. **Real Estate MFs** — properties (rent + appreciation) and real-estate company securities; must be **close-ended and listed**.
- → **Thematic** — trends/themes (multi-sector, international, commodity); broader than sector; market may take long to recognise the theme.
- → **Arbitrage** — open-ended equity scheme; mix of cash equities, equity derivatives, debt; captures spot–futures price differential; better than debt, less volatile than equity.
- → **Hedge fund** — lightly regulated private vehicle for selected clients; reveals nothing publicly; charges a **performance fee** (cut of profits above benchmark; losses borne by investors); leveraged, long, short, derivative positions; aim is to **maximise return** (not to hedge).
- → **Cash fund** — open-ended liquid scheme; very short-dated debt and money market; growth and dividend options.

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TYK P22 (ANP Plan, hedge fund ₹ 20 crore; fee 0.10% of assets + 2% incentive on value above the maximum value since inception ₹ 21 crore): fixed fee 0.10% × 20 cr = ₹ 2,00,000 in all cases. (a) 29%: 1.29 × 20 = 25.80 cr; excess over 21 cr = 4.80 cr × 2% = 9,60,000 → total **₹ 11,60,000**. (b) 4.5%: 20.90 cr < 21 cr → no incentive → **₹ 2,00,000**. (c) −1.8%: **₹ 2,00,000**.

3.9 Exchange Traded Funds

→ Hybrid of **listed stock + index fund**; listed, price linked to underlying index; **authorised participants** act as market makers.

→ Bought/sold **any time in market hours** at real-time prices close to NAV (open-ended schemes deal at end-of-day NAV); no paperwork, order through broker; low cost, tax efficient, stock-like.

→ Combines MF valuation feature (end-of-day NAV) with close-ended fund tradability (intra-day, at more or less than NAV).

→ Types: **Index ETFs** (replicate index); **Commodity ETFs** (precious metals, futures); **Bond ETFs** (thrive in recessions — money moves from stocks to bonds; indicate economic conditions); **Currency ETFs** (total return: FX spot change, local institutional interest rates, collateral yield).

3.10 Fixed Maturity Plans

→ **Close-ended**, invest at NFO; CDs, CPs, money market, NCDs, sometimes bank FDs, over a fixed period; listed but thinly traded → **not liquid**.

→ **Free from interest rate risk** (instruments mature with the fund) but carry **credit risk** (downgrade cuts return).

→ Mostly 3-year tenure for **indexation** benefit; to earn more, part goes into **AA and below** rated debt (higher coupon).

4 Advantages of Mutual Fund (also TYK T2)

→ **Professional management** — skilled managers + research analysts.

→ **Diversification** — reduces concentration risk.

→ **Convenient administration** — demat, no share-transfer risk, no delay.

→ **Higher returns** over medium to long term (judge by long-run average, not boom-phase returns).

→ **Low cost of management** — SEBI cap **2.50%** for equity MFs; excess borne by AMC.

→ **Liquidity** — open-ended: sale/repurchase by MF; close-ended: listing.

→ **Transparency** — portfolio disclosed at least **half-yearly** (many quarterly/monthly); NAV daily for open-ended, published through AMFI.

→ **Other benefits** — SWP, SIP, switching (except tax saving fund: no switch-out for 3 years).

→ **Highly regulated** — SEBI registration, MF Regulations.

→ **Economies of scale** — pooled money is cheaper; retail access to capital market.

→ **Flexibility** — SIP, SWP, wide range of schemes. **Convenience** — easy via digital portals.

5 Drawbacks of Mutual Fund

→ **No guarantee of return** — (i) some schemes underperform benchmark; (ii) fund gain ≠ every investor's gain (different entry/exit points); (iii) massive fall can erode principal short-term (2008 crisis, Covid 2020); remote over long term.

→ **Diversification** — minimises risk but does not maximise return; one stock doubling doesn't double the fund; neither exceptionally good nor poor.

→ **Selection of proper fund** — harder than picking a share; past performance can't predict future.

→ **Cost factor** — fund management fees in annual recurring expenses; **exit load** if withdrawn before a stipulated period; AMC fee not related to performance.

6, 6.1 Terms — Net Asset Value (NAV)

Asked: Sep 2025 MCQ CS-11 4-6 6m

Asked: Jan 2026 Q6(a) 6m

→ NAV = value per unit of portfolio holding = what a unit holder would get per unit if the fund were redeemed. Returns depend on **NAV** and **costs**.

→ Three aspects: (i) **net** value — market value of assets less liabilities; (ii) **changes daily**; (iii) **per unit** — ÷ outstanding units.

→ Investors' subscription = **unit capital** (liability side); investments = assets.

Formula	Terms / logic	SM figures
NAV per unit = Net assets of the scheme ÷ Number of units outstanding	Per-unit claim of a holder.	TYK P7 (Cinderella, Scheme Rudolf): shares 4,20,20,000 − accrued exp 2,50,000 − other liab 2,00,000 = 4,15,70,000 ÷ 10,00,000 = **₹ 41.57**
Net assets = MV of investments + Receivables + Other accrued income + Other assets − Accrued expenses − Other payables − Other liabilities	Everything at **market value**; accrued (unpaid) items count.	
New units issued = Amount invested ÷ NAV at that time	New money enters at current NAV, so existing NAV is undisturbed; cash not invested stays as a bank asset.	TYK P8, P9

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TYK P8 (L, M, N, P Ltd; 6,00,000 units): (i) 4,00,000 + 93,72,000 + 72,24,000 + 3,03,06,000 = 4,73,02,000 ÷ 6,00,000 = ₹ 78.8366 → **₹ 78.84**. (ii) Mr. A ₹ 30,00,000; 8,000 M Ltd bought (M now 38,000 = 1,18,71,200); cash 5,00,800; total 5,03,02,000; units = 6,00,000 + 30,00,000 ÷ 78.8366 = **6,38,053**. (iii) 2 Feb: 4,10,000 + 1,36,80,000 + 76,62,000 + 3,02,34,000 + 5,00,800 = 5,24,86,800 ÷ 6,38,053 = **₹ 82.26**.

TYK P9 (Fair Return MF; 8,00,000 units): (a) 9,52,38,000 ÷ 8,00,000 = ₹ 119.0475 → **₹ 119.05**. (b) Mr. X ₹ 50,00,000; 18,000 C Ltd bought (C now 28,000 = 74,03,200); cash 2,40,800; total 10,02,38,000; units = 8,00,000 + 50,00,000 ÷ 119.0475 = **8,42,000**. (c) 2 April: 10,27,87,200 ÷ 8,42,000 = **₹ 122.075**.

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TYK P18 (fund flow → NAV; 10,00,000 units of ₹ 10, investments 98,00,000): opening cash 100 − 98 lakh = 2,00,000 + dividend 12,00,000 + interest 56,000 (7% G-sec) + 45,000 (9% deb) + 50,000 (10% deb) = 15,51,000 − opex 5,00,000 = cash **10,51,000**. Assets: cash 10,51,000 + G-sec 8,00,000 (par) + 50,000 shares @ 175 = 87,50,000 + 9% unlisted deb 5,00,000 (cost) + 10% deb @ 90% = 4,50,000 = 1,15,51,000 → NAV **₹ 11.55**. If ₹ 0.80/unit dividend paid: 1,15,51,000 − 8,00,000 = 1,07,51,000 → **₹ 10.75**.

TYK P19 (regular income scheme; 20 lakh units): listed shares 20 × 2,300 ÷ 1,000 = 46.00; cash 1.23; unlisted bonds 1 × 80% = 0.80; listed bonds at NAV-date value 8.00; dividend accrued 0.80; other fixed-interest at cost 4.50 → assets 61.33 cr − (payable on shares 6.32 + exp accrued 0.75 = 7.07) = 54.26 cr ÷ 20,00,000 = **₹ 271.30**.

TYK P15 (ABC MF, 20 lakh units @ ₹ 10, initial exp 12 lakh, portfolio 185 lakh): opening bank 200 − 185 − 12 = 3.00 + sale 63 + dividend 2 = 68 − purchases 56 − expenses paid 90% × 8 = 7.20 − capital gain distributed 80% × (63 − 60) = 2.40 − dividend distributed 80% × 2 = 1.60 → closing bank 0.80. Net assets 198 + 0.80 − arrears 0.80 = 198 → NAV **₹ 9.90**. Per unit: income received (2.40 + 1.60) ÷ 20 = 0.20, less fall in value (200 − 198) ÷ 20 = 0.10 → 0.10 on ₹ 10 = **1% monthly, 12% annual**.

6.1 (contd.) Return on MF units — formulas used in ICAI answers

Asked: Sep 2025 Q3(b) 4m

Asked: Sep 2025 Q5(b) 7m

Asked: Jan 2026 Q1(a) 6m

Asked: May 2026 Q3(b) 4m

Formula	Terms / logic	SM figures
Return = $[(NAV_1 - NAV_0) + D_1 + CG_1] \div NAV_0 \times 100$	Distributions taken in **cash**: change in NAV + dividend (income) + capital gain distribution, per unit, over opening NAV.	TYK P4: (13 − 12.25) + 1.25 + 1.00 = 3.00 ÷ 12.25 = **24.49%**. P5: 0.60 + 0.90 + 0.75 = 2.25 ÷ 8.50 = **26.47%**. P14: 0.70 + 0.75 + 0.60 = 2.05 ÷ 8.75 = **23.43%**.
$r = [(NAV_t - NAV_{t-1}) + I_t + G_t] \div NAV_{t-1}$	Same, for one period (month); I_t income, G_t capital gain distribution. Annual = monthly × 12.	TYK P3: (0.06 + 0.0375 + 0.03) ÷ 20 = 0.1275 ÷ 20 = **0.6375% p.m. = 7.65% p.a.** TYK P6: 15% p.a. → 1.25% p.m.; 0.0125 = (NAV_t − 65.78 + 0.50 + 0.32) ÷ 65.78 → NAV_t = **₹ 65.78** (no change).
Additional units = Total amount reinvested ÷ reinvestment price; Return = (Value of all units at year end − Price paid for original units) ÷ Price paid for original units	Distributions **reinvested**: compare closing value of all units with original cost. *(app-written general form of ICAI's working)*	P4: 2.25 × 300 = 675 ÷ 12.50 = 54; 354 units × 13 = 4,602 vs 3,675 → **25.22%**. P5: 1.65 × 200 = 330 ÷ 8.75 = 37.71; 237.71 units = 2,163.16 vs 1,700 → **27.24%**. P14: 1.35 × 300 = 405 ÷ 8.65 = 46.82; 346.82 × 9.45 = 3,277.45 vs 2,625 → **24.85%** → reinvestment preferable.
Effective yield p.a. = h ÷ a × 365 ÷ i × 100	ICAI table: a = investment; c = units (a ÷ b, b = opening NAV); e = total NAV at end (c × d); f = e − a; g = dividend received; h = total yield (f + g); i = number of days. Units = investment ÷ entry NAV. Days counted including entry date (1.12 to 31.3 = 121).	TYK P10 (Mr. Sinha): X 47,619.05 units × 10.40 = 4,95,238.12; yield (−4,761.88 + 9,500) = 4,738.12 over 121 days → **2.859%**; Y 2,500 over 90 days → **10.139%**; Z −1,000 over 31 days → **(−) 23.55%**.
Period of holding (days) = Yield (%) × 365 ÷ Effective yield p.a. (%)	Reverse of the above (ICAI working, TYK P11).	TYK P11 (Mr. Y): units 19,417.48 / 39,603.96 / 20,000; total NAV 7,99,068.77; yields 5,029.17 / −3,960.40 / 9,000; total yield 10,068.77 ÷ 8,00,000 = 1.2586%; yield % 2.5146 / −0.9901 / 4.5 → **95 / 31 / 68 days** → invested **27.12.17 / 01.03.18 / 23.01.18** (alternative 26.12.17 / 28.02.18 / 22.01.18).

- **Dividend reinvestment plan:** dividend = units held × **face value ₹ 10** × dividend %; new units = dividend ÷ NAV on that date (ex-dividend). Keep a running unit balance.
- **Bonus plan:** ratio a : b → a bonus units for every b held (5 : 4 on 10,000 = 12,500). No cash; value = final units × final NAV.
- **Annual (average) return** = Net gain ÷ Investment × 12 ÷ months held × 100 (simple). Alternative **IRR** = LR + [NPV at LR ÷ (NPV at LR − NPV at HR)] × (HR − LR).
- Annualised yield → yield for the period = annualised yield × months ÷ 12 (ICAI: 153.33 × 9 ÷ 12 = 115%; used to back-solve NAVs).

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TYK P17 (Mr. K, ₹ 2,00,000 each, 1.4.2005 to 31.3.2010 = 5 years): Plan D: 2,00,000 ÷ 38.20 = 5,235.60 units; dividends reinvested: 10% → 5,235.60 ÷ 39.10 = 133.90 (5,369.50); 15% → 8,054.25 ÷ 44.20 = 182.22 (5,551.72); 13% → 7,217.24 ÷ 45.05 = 160.20 (5,711.92); 16% → 9,139.07 ÷ 44.80 = 204 (5,915.92); maturity 40.40 × 5,915.92 = 2,39,003.17; gain 39,003.17 ÷ 2,00,000 ÷ 5 = **3.90%** (IRR: NPV 4% −3,557, 2% 16,473 → **3.645%**). **Plan B**: 2,00,000 ÷ 35.60 = 5,617.98; 1:5 → 1,123.60 (6,741.58); 1:8 → 842.70 (7,584.28); 1:10 → 758.43 (8,342.71); × 39.70 = 3,31,205.59; gain 1,31,205.59 → **13.12%** (IRR: NPV 13% −20,235, 8% 25,413 → **10.78%**).

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TYK P16 (Chanakya Opportunity Fund; ₹ 1,00,000 each at ₹ 10 on 1.4.1999; redeemed 31.07.2009 = 124 months; STT 0.2% on redemption; STCG 10% on units held under a year, LTCG exempt):

- **Mrs. Charu, Plan A (reinvestment)**: 10,000 → 20% 20,000 ÷ 30.70 = 651.47 (10,651.47) → 70% 74,560.29 ÷ 58.42 = 1,276.28 (11,927.75) → 40% 47,711.00 ÷ 42.18 = 1,131.13 (13,058.88) → 25% 32,647.20 ÷ 46.45 = 702.85 (13,761.73) → 40% 55,046.92 ÷ 48.10 = 1,144.43 (**14,906.16**). × 53.75 = 8,01,206.10 − STT 1,602.41 = 7,99,603.69 − STCG 10% on 1,144.43 × (53.64 − 48.10) = 6,340 → 634 → 7,98,969.69; gain 6,98,969.69 × 12 ÷ 124 → **67.64%**.
- **Mr. Anand, Plan B (bonus)**: 10,000 + 12,500 (5:4) = 22,500 + 7,500 (1:3) = 30,000 + 7,500 (1:4) = **37,500** × 22.98 = 8,61,750 − STT 1,723.50 = 8,60,026.50 − STCG 10% on 7,500 × (22.93 − 19.95) = 22,350 → 2,235 → 8,57,791.50; gain 7,57,791.50 → **73.33%**.
- **Mr. Bacchan, Plan C (growth)**: 10,000 × 82.07 = 8,20,700 − STT 1,641.40 = 8,19,058.60; no STCG; gain 7,19,058.60 → **69.59%**.

ICAI: 53.64 and 22.93 are NAV net of 0.2% STT; alternatively take NAV without STT (Proviso 5 to s. 48: STT not deductible in computing capital gain); the cost of the last bonus units "can also be taken as zero".

Note (app-written): ICAI's answer dates the 40% dividend "30.10.2007"; the question table says 31.10.2007 (SM p. 8.30 vs 8.46). Figures unaffected.

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TYK P12 (Mr. X, 10,000 units of ₹ 10 on 1.7.2007, dividends reinvested): 9-month yield = 153.33 × 9 ÷ 12 = 115% → value 31.3.2008 = 1,00,000 + 1,15,000 = 2,15,000; NAV = (2,15,000 − dividend 10,000) ÷ 10,000 = **₹ 20.50**; units = 10,000 + 10,000 ÷ 20.50 (487.80) = 10,487.80. Dividend 2009 = 10,487.80 × 10 × 20% = 20,975.60; 11,296.11 = 20,975.60 ÷ X + 10,487.80 → NAV 31.3.2009 = **₹ 25.95**. NAV 31.3.2010 = 1,00,000 (1 + 0.7352 × 33 ÷ 12) ÷ 11,296.11 = **₹ 26.75**.

TYK P13 (same data as HPR): HPR 115% → NAV 31.3.2013 **₹ 20.50**; NAV 31.3.2014 **₹ 25.95**; NAV 31.3.2015 = 1,00,000 (1 + 2.0217) ÷ 11,296.11 = **₹ 26.75**.

6.2 Entry and Exit Load

Asked: Sep 2025 Q5(b) 7m | Asked: May 2026 Q3(b) 4m

- Loads = sales charges to compensate **distribution costs**; funds without them = **no-load funds**.
- **Entry load** — at purchase; **% added to NAV** → fewer units allotted. At present no scheme has an entry load.
- **Exit load** — at redemption or switch; **% deducted from NAV**; goes to the **scheme's pool**, not the AMC. Most schemes have none after a stipulated holding period.

Formula	Terms / logic	SM figures
$r_2 = r_1 \div (1 - \text{initial exp}) + \text{recurring exp.}$	(TYK PI answer writes the same as $R_2 = R_1 \div (1 - \text{Initial expenses \%}) + \text{Recurring expenses \%.}$) r_1 = return investor earns on his own; r_2 = return the fund must earn; initial expenses reduce the amount actually invested, so the fund must earn r_1 on the smaller base, plus cover annual recurring expenses.	SM Example (Mr. X): 0.10 ÷ (1 − 0.06) + 0.02 = **12.64%**. TYK PI (Mr. A): 16% ÷ (1 − 0.055) + 1.5% = **18.43%**.
Public Offer Price = NAV ÷ (1 − Front end Load); Redemption Price = NAV ÷ (1 + Back End Load)	ICAI's main method.	TYK P2 (ELSS NAV ₹ 10, POP ₹ 10.204, redemption ₹ 9.80): main — front-end 0.0199 say **2%**, back-end **2.04%**; alternative — front-end (10.204 − 10) ÷ 10 = **2.04%**, exit (10 − 9.80) ÷ 10 = **2.00%**.
Alternative: Front End Load = (Public Offer Price − NAV) ÷ NAV; Exit Load = (NAV − Redemption Price) ÷ NAV	Load as % of NAV (app-written general form of ICAI's alternative working).	
Issue price = Opening NAV + Entry Load + Dividend Equalization; Repurchase price = Opening NAV − Exit Load + Dividend Equalization	(app-written general form of ICAI's TYK P20 working.) Equalisation = income per unit accumulated up to that date, collected from new holders / paid to leavers so income per unit for others is unchanged.	TYK P20 below.

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TYK P20 (open-ended, 300 lakh units, NAV ₹ 18.75): income per unit April 22.95 ÷ 300 = 0.0765; equalisation on 6 lakh issued 0.4590 → 23.4090 on 306; May 0.1125 (34.425) → 0.1890 / 57.8340; equalisation paid on 3 lakh repurchased (0.5670) → 57.2670 on 303; June 0.1500 (45.45) → **(i) available 102.7170 lakh = ₹ 0.3390/unit**; 70% paid 0.2373 = 71.9019; retained 30.8151. **(ii) Issue price** 18.750 + 0.375 + 0.0765 = **₹ 19.2015**. **(iii) Repurchase** 18.750 − 0.375 + 0.1890 = **₹ 18.564**. **(iv)** 5,625 + appreciation 425.47 + issue 6 × 19.2015 = 115.209 + income 102.825 = 6,268.504 − repurchase 3 × 18.564 = 55.692 − distributed 71.9019 = 6,140.9101 ÷ 303 lakh = **₹ 20.2670**.

Note (app-written): ICAI's repurchase working labels the 0.1890 line "Add: Dividend Equalization paid on Issue Price"; it is the equalisation on repurchase (SM p. 8.51). Figures unaffected.

6.3–6.4 Trail commission; Expense ratio

- **Trail commission** — paid by investor to advisor **each year** as incentive to review holdings; usually **0.30–0.75%** of investment value; on **daily average AUM** of the distributor, paid **monthly**; separate from upfront commission paid by the fund company out of its own pocket.
- **Expense ratio** (MER / TER) — % of assets spent to run the fund: management and advisory fees, travel, audit, custodian, registrar, consultancy fees. **Excludes brokerage** on portfolio trading. Can be 2–3% — can seriously undermine performance.

6.5 Side Pocketing

- Separation of **risky/doubtful (stressed, illiquid) assets** from other investments and cash; money linked to stressed assets is locked until recovered; avoids **distress selling**.
- When rating falls, illiquid assets moved to a side pocket → NAV reflects actual value of **liquid assets**; benefits current long-term holders; liquidity not a problem despite frequent allotments/redemptions.
- Common internationally, but also misused to deprive investors of genuine returns. India: **IL&FS** default — MFs wrote off exposure, low to negative returns.

6.6 Tracking Error

→ Divergence of a fund's return from its **benchmark** return — passive managers often can't replicate exactly. Computed on benchmark return vs quarterly or monthly average NAVs.

→ **Higher TE → higher risk profile**; shows benchmark not followed properly. Ideal ≈ **zero**.

→ Reasons: **transaction cost, AMC fees, fund expenses, cash holdings, sampling biasness**.

Formula	Terms / logic
$TE = \sqrt{[\ \Sigma(d - \bar{d})^2 \div (n - 1)\ ]}$	SM: d = differential return; $\bar{d}$ = average differential return; n = no. of observations. (app-written: d is fund return − benchmark return each period; TE is the standard deviation of these differences.)

7, 7.1 Evaluation of Mutual Funds; Quantitative parameters

→ Purpose: fund keeps generating maximum profit with minimum risk; decide hold or replace; past performance doesn't guarantee future. Not daily; give equity funds **3–5 years**; review at least every **6/12 months**.

→ **Risk-adjusted returns** — return relative to risk; same return → choose lower risk.

→ **Benchmark returns** — excess over benchmark = **alpha** (index 11%, fund 13% → outperformed; 8% → underperformed).

→ **Comparison to peers** — same category ("apples with apples"); should consistently beat peers.

→ **Returns across economic and market cycles** — not just 6/12-month windows.

→ **Financial measures**: **expense ratio**; **Sharpe** = $[E(R) - R_f] \div \sigma$ (total risk); **Treynor** = $[E(R) - R_f] \div \beta$ (systematic risk; fund beta = volatility vs its benchmark); **Sortino** — Sharpe with **downside deviation** instead of total SD.

ICAI ·

TYK P21 (D MF NAV ₹ 70.71, 99% equity; K MF ₹ 62.50, 96% equity; market −5% in a month; cost ₹ 3 p.a. per unit): equity/cash D 70 / 0.71, K 60 / 2.50. D: $E(R) - R_f = 2 \times 11.25 = 22.50$, $\beta_D = 22.50 \div 15 =$ **1.50**; K: $3.3 \times 5 = 16.50$, $\beta_K = 16.50 \div 15 =$ **1.10**. Equity falls 7.50% / 5.50%; cash less $3 \div 12 = 0.25$ → 0.46 / 2.25. NAV: D 64.75 + 0.46 = **₹ 65.21**; K 56.70 + 2.25 = **₹ 58.95**.

TYK P23 (Ms. Sunidhi, Constant Ratio Plan — formula plan from the Portfolio Management chapter: keep equity fund : bonds at the starting ratio, here 1 : 1, and rebalance when the trigger is hit): ₹ 2 lakh → 1,00,000 bonds + 2,500 units @ 40. NAV 25: total 1,62,500 → 81,250 each, **buy 750** (3,250 units). NAV 36: 1,98,250 → **sell 496.53** (2,753.47). NAV 38: 2,03,756.86 → **sell 72.46** (2,681.01). NAV 50: 2,35,928.93 → **sell 321.72** (2,359.29). NAV 52: ending value **₹ 2,40,647.55** vs buy and hold **₹ 2,60,000**.

Note (app-written): ICAI rebalances when NAV moves 15% or more from the previous month's NAV (25→36, 32→38, 43→50). ICAI's buy-and-hold ₹ 2,60,000 = the whole ₹ 2 lakh in the fund (5,000 units × 52); a 50 : 50 buy and hold would be 1,00,000 + 2,500 × 52 = ₹ 2,30,000 (SM p. 8.54).

7.2 Qualitative parameters

→ **Quality of portfolio** — equity: blue-chip/large companies, diversification, style (growth, value, blend); debt: credit quality, average maturity, modified duration; investments as per fund objective; **lower portfolio turnover** ratio is better.

→ **Track record and competence of fund manager** — knowledge, ability, past performance.

→ **Credibility of fund house team** — redemptions, crediting dividends, information.

8, 8.1–8.2 Role of Fund Managers — Active and Passive funds

→ ***Actively managed*** — *research, judgement, due diligence to* ***outperform the market*** *and generate* ***positive alpha*** *(stock picking).*

→ ***Passively managed*** — *match the index return with* ***minimum tracking error.***

→ *Other roles:* ***Compliances*** *(SEBI, AMFI guidelines; reporting expense ratio, redemptions; investor awareness; documents on time);* ***constant monitoring*** *of performance (entry/exit decisions; growth above inflation and interest rate);* ***creation and protection of wealth*** *(no reckless risk; fundamental and technical analysis);* ***control over outsourced work*** *to third parties.*

9 Role of FIIs in Mutual Funds

→ *FIIs = large foreign groups with substantial investible funds, registered abroad; strong research teams; fuel bullish markets for short periods → strong forex inflow. Buy units of domestic MFs* ***directly from the issuer*** *or* ***through a registered broker*** *on a recognised exchange, within* ***SEBI limits.***

→ *Importance:* ***enhanced corporate governance*** *(due diligence),* ***improved competition*** *in market,* ***improved capital inflow*** *(employment, forex position).*

Exam steps — NAV: *1 each holding × market price (index-adjust shares bought at old index; impair unlisted) → 2 + cash (opening cash + income − expenses paid − distributions) + accrued income → 3 − accrued expenses and payables → 4 ÷ units (new money: units = amount ÷ NAV at that time).*

Exam steps — HPR payout vs reinvestment: *1 cash: $[(NAV_1 - NAV_0) + D_1 + CG_1] \div NAV_0$ → 2 reinvest: $(D_1 + CG_1) \times$ units ÷ reinvestment NAV = additional units → 3 (value of all units at NAV_1 − price paid) ÷ price paid → 4 pick higher %. With loads: buy at NAV + entry load, sell at NAV − exit load; then tax, then real return.*

Exam steps — Dividend vs Bonus plan: *1 units = investment ÷ entry NAV → 2 dividend plan: units × ₹ 10 × % ÷ that day's NAV, running total → 3 bonus plan: units × ratio, running total → 4 final units × final NAV (− STT, − STCG on units under a year) → 5 gain ÷ investment × 12 ÷ months (or IRR).*

Exam steps — back-solve NAV (TYK P12/13): *1 period HPR (annual × months ÷ 12) → 2 value = investment × (1 + HPR) → 3 NAV = (value − dividend) ÷ units → 4 dividend ÷ NAV = new units → 5 next NAV = dividend ÷ (final units − previous units).*

Exam steps — issue/repurchase with equalisation (TYK P20): *1 income per unit month by month on units then outstanding → 2 issue = NAV + load + accumulated income/unit; repurchase = NAV − load + accumulated income/unit → 3 closing net assets = opening + appreciation + issue proceeds + income − repurchases − distribution → 4 ÷ closing units.*

Exam steps — expected NAV (TYK P21): 1 split NAV into equity and cash → 2 $E(R) - R_f$ = Sharpe × σ, then β = [$E(R) - R_f$] ÷ Treynor → 3 equity × (1 − β × market fall) → 4 cash − monthly cost → 5 add.

Past papers

⑧ Mutual Funds — what ICAI asked (Sep 2025, Jan 2026, May 2026)

37 marks *set across the last three attempts (10.5% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	MCQ CS-II 4-6	6	Mutual fund accounts: closing net cash balance, total NAV, and NAV per unit after a dividend per unit is distributed (MCQ)
Sep 2025	Q3(b)	4	Mutual fund holding period return: payout of dividend and capital gain versus reinvestment at average NAV; choose the better option (Numerical)
Sep 2025	Q5(b)	7	Mutual fund: units bought after entry load, NAV after appreciation, distribution reinvested at year-end NAV, return and real return after exit load, tax and inflation (Numerical)
Jan 2026	Q1(a)	6	Dividend (bonus) scheme: back-solve opening NAV per unit from average annual yield, two bonus issues (1:4 and 2:5) and closing NAV (Numerical)
Jan 2026	Q6(a)	6	New fund offer: closing NAV per unit after underwriting commission, marketing cost, gains, regulated expenses, income distribution policy (Numerical)
May 2026	Q3(b)	4	Dividend plan versus bonus plan of a mutual fund: total return and holding period return after entry/exit loads; which plan is better (Numerical)
May 2026	Q4(b)	4	Closed-ended fund: market price from NAV and premium/discount, total, average and geometric return per period for the investor, geometric growth of NAV, interpretation (Numerical)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Explain how to establish a Mutual Fund.

ICAI answer

1. Establishment of a Mutual Fund: A mutual fund is required to be registered with the Securities and Exchange Board of India (SEBI) before it can collect funds from the public. All mutual funds are governed by the same set of regulations and are subject to monitoring and inspections by the SEBI. The Mutual Fund has to be established through the medium of a sponsor. A sponsor means any body corporate which, acting alone or in combination with another body corporate, establishes a mutual fund after completing the formalities prescribed in the SEBI's Mutual Fund Regulations.

The role of sponsor is akin to that of a promoter of a company, who provides the initial capital and appoints the trustees. The sponsor should be a body corporate in the business of financial services for a period not less than 5 years, be financially sound and be a fit party to act as sponsor in the eyes of SEBI.

The Mutual Fund has to be established as either a trustee company or a Trust, under the Indian Trust Act and the instrument of trust shall be in the form of a deed. The deed shall be executed by the sponsor in favour of the trustees named in the instrument of trust. The trust deed shall be duly registered under the provisions of the Indian Registration Act, 1908. The trust deed shall contain clauses specified in the Third Schedule of the Regulations.

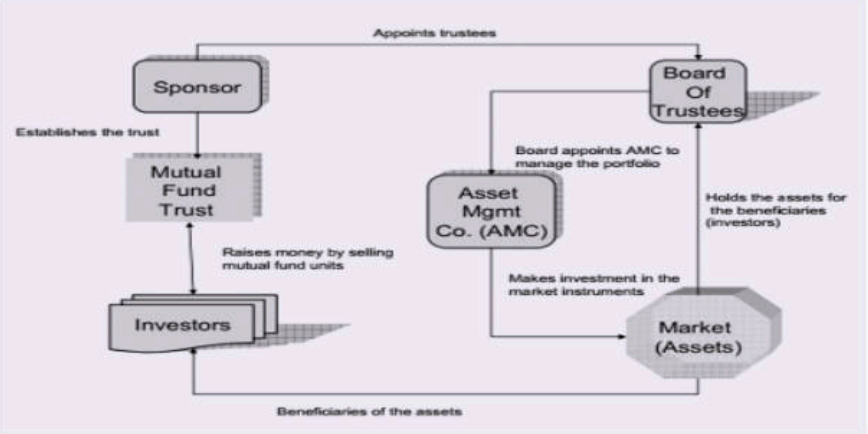
An Asset Management Company, who holds an approval from SEBI, is to be appointed to manage the affairs of the Mutual Fund and it should operate the schemes of such fund. The Asset Management Company is set up as a limited liability company, with a minimum net worth of 100 crores.

The sponsor should contribute at least 40% to the net worth of the Asset Management Company. The Trustee should hold the property of the Mutual Fund in trust for the benefit of the unit holders.

SEBI regulations require that at least two-thirds of the directors of the Trustee Company or board of trustees must be independent, that is, they should not be associated with the sponsors. Also, 50 per cent of the directors of AMC must be independent. The appointment of the AMC can be terminated by majority of the trustees or by 75% of the unit holders of the concerned scheme.

The AMC may charge the mutual fund with Investment Management and Advisory fees subject to prescribed ceiling. Additionally, the AMC may get the expenses on operation of the mutual fund reimbursed from the concerned scheme.

The Mutual fund also appoints a custodian, holding valid certificate of registration issued by SEBI, to have custody of securities held by the mutual fund under different schemes. In case of dematerialized securities, this is done by Depository Participant. The custodian must be independent of the sponsor and the AMC.



Theory Q2

2. What are the advantages of investing in Mutual Funds?

ICAI answer: "Please refer paragraph 4" — SM paragraph 4 reproduced

4. ADVANTAGES OF MUTUAL FUND

(a) **Professional Management:** The funds are managed by skilled and professionally experienced managers backed by a team of Research Analysts.

(b) **Diversification:** Mutual Funds invest into many securities and offer diversification which reduces the concentration risk.

(c) **Convenient Administration:** There are no administrative risks of share transfer, as many of the Mutual Funds offer services in a demat form which saves investor's time and prevents delay.

(d) **Higher Returns:** Over a medium to long-term investment horizon, investors get higher returns in Mutual Funds as compared to other avenues of investment. However, investors are cautioned that such very high returns during the exceptional bull phase of the market like IT

boom or Infrastructure boom should not be considered as regular returns and therefore one should look at the average returns provided by the Mutual Funds particularly in the equity schemes over a long period of time.

(e) **Low Cost of Management:** SEBI has prescribed maximum limit of charging 2.50% for Equity Mutual Funds. No Mutual Fund can increase the cost beyond prescribed limits of 2.5% maximum and any extra cost of management is to be borne by the AMC.

(f) **Liquidity:** In all the open ended funds, liquidity is provided by direct sales / repurchase by the Mutual Fund and in case of close ended funds, the liquidity is provided by listing the units on the Stock Exchange.

(g) **Transparency:** The SEBI Regulations now compel all the Mutual Funds to disclose their portfolios on a half-yearly basis. However, many Mutual Funds disclose their Scheme Portfolio on a quarterly or monthly basis to their investors. The NAVs are calculated on a daily basis in case of open ended funds and are published through AMFI in the newspapers.

(h) **Other Benefits:** Mutual Funds provide systematic withdrawal and systematic investment plans according to the need of the investors. The investors can also switch from one scheme to another without any restrictions except in case of Tax Savings Fund which restricts switch out for first 3 years of its investments.

(i) **Highly Regulated:** Mutual Funds all over the world are highly regulated and in India all Mutual Funds are registered with SEBI and are strictly regulated as per the Mutual Fund Regulations which provide high level of investor protection.

(j) **Economies of scale:** The way mutual funds are structured gives it a natural advantage. The "pooled" money from numerous investors ensures that mutual funds enjoy economies of scale; it is cheaper compared to investing directly in the capital markets which involves higher charges. This also allows retail investors access to participation in the Capital Market which otherwise is difficult for them go directly.

(k) **Flexibility:** One of the biggest advantages of a Mutual Fund Scheme is its flexibility. An investor can opt for Systematic Investment Plan (SIP), Systematic Withdrawal Plan etc. to plan his cash flow requirements as per his convenience. The wide range of schemes being launched in India by different mutual funds also provides an added flexibility to the investor to plan his portfolio accordingly.

(l) **Convenience:** It is very convenient & easy to invest & disinvest from Mutual Fund Schemes specially through digital transaction portals.

Theory — TYK with ICAI answers

ICAI answer: "Please refer paragraph 4" — SM paragraph 4 reproduced

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Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical Q1

1. Mr. A can earn a return of 16 per cent by investing in equity shares on his own. Now he is considering a recently announced equity based mutual fund scheme in which initial expenses are 5.5 per cent and annual recurring expenses are 1.5 per cent. How much should the mutual fund earn to provide Mr. A return of 16 per cent?

Solution — ICAI

1. Personal earnings of Mr. A = R_1 = 16%

Mutual Fund earnings = R_2

$$R_2 = \frac{1}{1 - \text{Initial expenses (\%)}} R_1 + \text{Recurring expenses (\%)}$$

$$= \frac{1}{1-0.055} \times 16\% + 1.5\%$$

= 18.43%

Mutual Fund earnings = 18.43%

Practical Q2

2. The unit price of Equity Linked Savings Scheme (ELSS) of a mutual fund is ₹ 10/-. The public offer price (POP) of the unit is ₹ 10.204 and the redemption price is ₹ 9.80.

Calculate:

(i) Front-end Load

(ii) Back end Load

Solution — ICAI

2. Public Offer Price = NAV/ (1 – Front end Load)

Public Offer Price: ₹ 10.204 and NAV: ₹ 10

Accordingly,

10.204 = 10/(1 – F)

F = 0.0199 say 2%

Redemption Price = NAV/ (1 + Back End Load)

₹ 9.80 = 10/ (1 + Back End Load)

B = 0.0204 i.e. 2.04%

Alternative

(i) Front End Load = $\frac{10.204 - 10.00}{10.00}$ = 0.0204 or 2.04%

(ii) Exit Load = $\frac{10.00 - 9.80}{10.00}$ = 0.020 or 2.00%

Practical Q3

3. A mutual fund that had a net asset value of ₹ 20 at the beginning of month - made income and capital gain distribution of ₹ 0.0375 and ₹ 0.03 per share respectively during the month, and then ended the month with a net asset value of ₹ 20.06. Calculate monthly return.

Solution — ICAI

3. Calculation of Monthly Return on the Mutual Funds

$$r = \left[\frac{(NAV_t - NAV_{t-1}) + I_t + G_t}{NAV_{t-1}}\right]$$

Where,

r = Return on the mutual fund

NAV_t = Net assets value at time period t

NAV_{t-1} = Net assets value at time period t – 1

I_t = Income at time period t

G_t = Capital gain distribution at time period t

$$= \left[\frac{(₹\ 20.06 - ₹\ 20.00) + (₹\ 0.0375 + ₹\ 0.03)}{20}\right]$$

$$= \frac{0.06 + 0.0675}{20}$$

$$= \frac{0.1275}{20} = 0.006375$$

Or, r = 0.6375% p.m.

Or = 7.65% p.a.

Practical Q4

4. An investor purchased 300 units of a Mutual Fund at ₹ 12.25 per unit on 31st December, 2009. As on 31st December, 2010 he has received ₹ 1.25 as dividend and ₹ 1.00 as capital gains distribution per unit.

Required:

(i) The return on the investment if the NAV as on 31st December, 2010 is ₹ 13.00.

(ii) The return on the investment as on 31st December, 2010 if all dividends and capital gains distributions are reinvested into additional units of the fund at ₹ 12.50 per unit.

Solution — ICAI

4. Return for the year (all changes on a per year basis)

Particulars	₹ /Unit
Change in price (₹ 13.00 – ₹ 12.25)	0.75
Dividend received	1.25
Capital gain distribution	1.00
Total Return	3.00

Return on investment = $\frac{3.00}{12.25} \times 100 = 24.49\%$

Alternatively, it can also be computed as follows:

$$\frac{(NAV_1 - NAV_0) + D_1 + CG_1}{NAV_0} X 100$$

$$= \frac{(13 - 12.25) + 1.25 + 1.00}{12.25} X 100$$

= 24.49%

If all dividends and capital gain are reinvested into additional units at ₹ 12.50 per unit the position would be.

Total amount reinvested = ₹ 2.25 × 300 = ₹ 675

Additional units added = $\frac{₹\ 675}{12.50}$ = 54 units

Value of 354 units as on 31-12-2010 = ₹ 4,602

Price paid for 300 units on 31-12-2009 (300 × ₹ 12.25) = ₹ 3,675

Return = $\frac{₹\ 4,602 - ₹\ 3,675}{₹\ 3,675} = \frac{₹\ 927}{₹\ 3,675}$ = 25.22%

Practical QS

5. SBI mutual fund has a NAV of ₹ 8.50 at the beginning of the year. At the end of the year NAV increases to ₹ 9.10. Meanwhile fund distributes ₹ 0.90 as dividend and ₹ 0.75 as capital gains.

   (i) What is the fund's return during the year?

   (ii) Had these distributions been re-invested at an average NAV of ₹ 8.75 assuming 200 units were purchased originally. What is the return?

Solution — ICAI

5. (i) Normal Return for the year (all changes on a per year basis)

Particulars	₹ /Unit
Change in price (₹ 9.10 – ₹ 8.50)	0.60
Dividend received	0.90
Capital gain distribution	0.75
Total Return	2.25

Return on investment = $\frac{2.25}{8.50} \times 100 = 26.47\%$

(ii) If all dividends and capital gain are reinvested into additional units at ₹ 8.75 per unit the position would be.

Total amount reinvested = ₹ 1.65 × 200 = ₹ 330

Additional units added = $\frac{₹\,330}{8.75}$ = 37.71 units

Value of 237.71 units at end of year = ₹ 2,163.16

Price paid for 200 units in beginning of the year (200 × ₹ 8.50) = ₹ 1,700

Return = $\frac{₹\ 2,163.16 - ₹\ 1,700}{₹\,1,700} = \frac{₹\,463.16}{₹\,1,700} = 27.24\%$

Practical Q6

6. The following information is extracted from Steady Mutual Fund's Scheme:

- Asset Value at the beginning of the month - ₹ 65.78
- Annualised return -15 %
- Distributions made in the nature of Income & Capital gain (per unit respectively). - ₹ 0.50 and ₹ 0.32

You are required to:

(i) Calculate the month end net asset value of the mutual fund scheme (limit your answers to two decimals).

(ii) Provide a brief comment on the month end NAV.

Solution — ICAI

6. (i) Calculation of NAV at the end of month:

Given Annual Return = 15%

Hence Monthly Return = 1.25% (r)

$$r = \frac{(NAV_t - NAV_{t-1}) + I_t + G_t}{NAV_{t-1}}$$

$$0.0125 = \frac{(NAV_t - ₹\ 65.78) + ₹\ 0.50 + ₹\ 0.32}{₹\ 65.78}$$

$0.82 = NAV_t - ₹\ 64.96$

$NAV_t = ₹\ 65.78$

(ii) There is no change in NAV.

Practical Q7

7. Cinderella Mutual Fund has the following assets in Scheme Rudolf at the close of business on 31st March,2014.

Company	No. of Shares	Market Price Per Share
Nairobi Ltd.	25000	₹ 20
Dakar Ltd.	35000	₹ 300
Senegal Ltd.	29000	₹ 380
Cairo Ltd.	40000	₹ 500

The total number of units of Scheme Rudolf are 10 lacs. The Scheme Rudolf has accrued expenses of ₹ 2,50,000 and other liabilities of ₹ 2,00,000. Calculate the NAV per unit of the Scheme Rudolf.

Solution — ICAI

7.

Shares	No. of shares	Price	Amount (₹)
Nairobi Ltd.	25,000	20.00	5,00,000
Dakar Ltd.	35,000	300.00	1,05,00,000
Senegal Ltd.	29,000	380.00	1,10,20,000
Cairo Ltd.	40,000	500.00	2,00,00,000
			4,20,20,000

Less: Accrued Expenses	2,50,000
Other Liabilities	2,00,000
Total Value	4,15,70,000
No. of Units	10,00,000
NAV per Unit (4,15,70,000/10,00,000)	41.57

Practical Q8

8. A Mutual Fund Co. has the following assets under it on the close of business as on:

Company	No. of Shares	1st February 2012 Market price per share ₹	2nd February 2012 Market price per share ₹
L Ltd	20,000	20.00	20.50
M Ltd	30,000	312.40	360.00
N Ltd	20,000	361.20	383.10
P Ltd	60,000	505.10	503.90

Total No. of Units 6,00,000

(i) Calculate Net Assets Value (NAV) of the Fund.

(ii) Following information is given:

Assuming one Mr. A, submits a cheque of ₹ 30,00,000 to the Mutual Fund and the Fund manager of this company purchases 8,000 shares of M Ltd; and the balance amount is held in Bank. In such a case, what would be the position of the Fund?

(iii) Find new NAV of the Fund as on 2nd February 2012.

Solution — ICAI

8. (i) NAV of the Fund

$$= \frac{₹\ 4,00,000 + ₹\ 93,72,000 + ₹\ 72,24,000 + ₹\ 3,03,06,000}{6,00,000}$$

$$= \frac{₹\ 4,73,02,000}{6,00,000} = ₹\ 78.8366 \text{ rounded to } ₹\ 78.84$$

(ii) The revised position of fund shall be as follows:

Shares	No. of shares	Price	Amount (₹)
L Ltd.	20,000	20.00	4,00,000
M Ltd.	38,000	312.40	1,18,71,200
N Ltd.	20,000	361.20	72,24,000
P Ltd.	60,000	505.10	3,03,06,000
Cash			5,00,800
			5,03,02,000

No. of units of fund = $6,00,000 + \frac{30,00,000}{78.8366} = 6,38,053$

(iii) On 2nd February 2012, the NAV of fund will be as follows:

Shares	No. of shares	Price	Amount (₹)
L Ltd.	20,000	20.50	4,10,000
M Ltd.	38,000	360.00	1,36,80,000
N Ltd.	20,000	383.10	76,62,000
P Ltd.	60,000	503.90	3,02,34,000
Cash			5,00,800
			5,24,86,800

NAV as on 2nd February 2012 = $\frac{₹\ 5,24,86,800}{6,38,053}$ = ₹ 82.26 per unit

Practical Q9

9. On 1st April 2009 Fair Return Mutual Fund has the following assets and prices at 4.00 p.m.

Shares	No. of Shares	Market Price Per Share (₹)
A Ltd.	10000	19.70
B Ltd.	50000	482.60
C Ltd.	10000	264.40
D Ltd.	100000	674.90
E Ltd.	30000	25.90
No. of units of funds		8,00,000

Please calculate:

(a) NAV of the Fund on 1st April 2009.

(b) Assuming that on 1st April 2009, Mr. X, a HNI, send a cheque of ₹ 50,00,000 to the Fund and Fund Manager immediately purchases 18000 shares of C Ltd. and balance is held in bank. Then what will be position of fund.

(c) Now suppose on 2 April 2009 at 4.00 p.m. the market price of shares is as follows:

Shares	₹
A Ltd.	20.30
B Ltd.	513.70
C Ltd.	290.80
D Ltd.	671.90
E Ltd.	44.20

Then what will be new NAV.

Solution — ICAI

9. (a) NAV of the Fund.

$$= \frac{₹\,1,97,000 + ₹\,2,41,30,000 + ₹\,26,44,000 + ₹\,6,74,90,000 + ₹\,7,77,000}{800000}$$

$$= \frac{₹\,9,52,38,000}{800000} = ₹\ 119.0475 \text{ rounded to } ₹\ 119.05$$

(b) The revised position of fund shall be as follows:

Shares	No. of shares	Price	Amount (₹)
A Ltd.	10000	19.70	1,97,000
B Ltd.	50000	482.60	2,41,30,000
C Ltd.	28000	264.40	74,03,200
D Ltd.	100000	674.90	674,90,000
E Ltd.	30000	25.90	7,77,000
Cash			2,40,800
			10,02,38,000

$$\text{No. of units of fund} = 800000 + \frac{5000000}{119.0475} = 842000$$

(c) On 2nd April 2009, the NAV of fund will be as follows:

Shares	No. of shares	Price	Amount (₹)
A Ltd.	10000	20.30	2,03,000
B Ltd.	50000	513.70	2,56,85,000
C Ltd.	28000	290.80	81,42,400
D Ltd.	100000	671.90	6,71,90,000
E Ltd.	30000	44.20	13,26,000
Cash			2,40,800
			10,27,87,200

$$\text{NAV as on 2nd April 2009} = \frac{₹\,10,27,87,200}{842000} = ₹\ 122.075 \text{ per unit}$$

Practical Q10

10. Mr. Sinha has invested in three Mutual fund schemes as per details below:

	Scheme X	Scheme Y	Scheme Z
Date of Investment	01.12.2008	01.01.2009	01.03.2009
Amount of Investment	₹ 5,00,000	₹ 1,00,000	₹ 50,000
Net Asset Value at entry date	₹ 10.50	₹ 10.00	₹ 10.00
Dividend received upto 31.03.2009	₹ 9,500	₹ 1,500	Nil
NAV as at 31.3.2009	₹ 10.40	₹ 10.10	₹ 9.80

You are required to calculate the effective yield on per annum basis in respect of each of the three schemes to Mr. Sinha upto 31.03.2009.

Solution — ICAI

10. Calculation of effective yield on per annum basis in respect of three mutual fund schemes to Mr. Sinha up to 31-03-2009:

Particulars		MF X	MF Y	MF Z
(a)	Investments	₹ 5,00,000	₹ 1,00,000	₹ 50,000
(b)	Opening NAV	₹10.50	₹10.00	₹10.00
(c)	No. of units (a/b)	47,619.05	10,000	5,000
(d)	Unit NAV ON 31-3-2009	₹ 10.40	₹ 10.10	₹ 9.80
(e)	Total NAV on 31-3-2009 (c x d)	₹ 4,95,238.12	₹ 1,01,000	₹ 49,000
(f)	Increase / Decrease of NAV (e - a)	(₹ 4,761.88)	₹ 1,000	(₹ 1,000)
(g)	Dividend Received	₹ 9,500	₹ 1,500	Nil
(h)	Total yield (f + g)	₹ 4,738.12	₹ 2,500	(₹ 1,000)
(i)	Number of Days	121	90	31
(j)	Effective yield p.a. (h/a x 365/i x 100)	2.859%	10.139%	(-) 23.55%

Practical QII

11. Mr. Y has invested in the three mutual funds (MF) as per the following details:

Particulars	MF 'X'	MF 'Y'	MF 'Z'
Amount of Investment (₹)	2,00,000	4,00,000	2,00,000
Net Assets Value (NAV) at the time of purchase (₹)	10.30	10.10	10
Dividend Received up to 31.03.2018 (₹)	6,000	0	5,000
NAV as on 31.03.2018 (₹)	10.25	10	10.20
Effective Yield per annum as on 31.03.2018 (percent)	9.66	-11.66	24.15

Assume 1 Year =365 days

Mr. Y has misplaced the documents of his investment. Help him in finding the date of his original investment after ascertaining the following:

(i) Number of units in each scheme;

(ii) Total NAV;

(iii) Total Yield; and

(iv) Number of days investment held.

Solution — ICAI

11. (i) Number of Units in each Scheme

MF 'X'	₹ 2,00,000 / ₹ 10.30	= 19,417.48
MF 'Y'	₹ 4,00,000 / ₹ 10.10	= 39,603.96
MF 'Z'	₹ 2,00,000 / ₹10.00	= 20,000.00

(ii) Total NAV on 31.03.2018

MF 'X'	= 19,417.48 x ₹ 10.25	₹ 1,99,029.17
MF 'Y'	= 39,603.96 x ₹ 10.00	₹ 3,96,039.60
MF 'Z'	= 20,000.00 x ₹ 10.20	₹ 2,04,000.00
Total		₹ 7,99,068.77

(iii) Total Yield

	Capital Yield	Dividend Yield	Total
MF 'X'	₹ 1,99,029.17- ₹ 2,00,000 = - ₹ 970.83	₹ 6,000	₹ 5,029.17
MF 'Y'	₹ 3,96,039.60 - ₹ 4,00,000 = - ₹ 3,960.40	Nil	- ₹ 3,960.40
MF 'Z'	₹ 2,04,000 - ₹ 2,00,000 = ₹ 4,000	₹ 5,000	₹ 9,000.00
Total			₹ 10,068.77

$$\text{Total Yield} = \frac{₹10,068.77}{₹\ 8,00,000} \times 100 = 1.2586\%$$

(iv) No. of Days Investment Held

	MF 'X'	MF 'Y'	MF 'Z'
Initial Investment (₹)	2,00,000	4,00,000	2,00,000
Yield (₹)	5,029.17	-3,960.40	9,000.00
Yield (%)	2.5146	- 0.9901	4.5
Period of Holding (Days)	$\frac{2.5146}{9.66} \times 365$ = 95 Days	$\frac{-0.9901}{-11.66} \times 365$ = 31 Days	$\frac{4.5}{24.15} \times 365$ = 68 Days
Date of Original Investment	27.12.17	01.03.18	23.01.18

Alternatively following dates can also be considered:

Date of Original Investment	26.12.17	28.02.18	22.01.18

Practical Q12

12. Mr. X on 1.7.2007, during the initial offer of some Mutual Fund invested in 10,000 units having face value of ₹ 10 for each unit. On 31.3.2008, the dividend paid by the M.F. was 10% and Mr. X found that his annualized yield was 153.33%. On 31.3.2009, 20% dividend was given.

On 31.3.2010, Mr. X redeemed all his balance of 11,296.11 units when his annualized yield was 73.52%. What are the NAVs as on 31.3.2008, 31.3.2009 and 31.3.2010?

Solution — ICAI

12. Yield for 9 months = (153.33 x 9/12) = 115%

Market value of Investments as on 31.03.2008 = 1,00,000/- + (1,00,000x 115%)

= ₹2,15,000/-

Therefore, NAV as on 31.03.2008 = (2,15,000-10,000)/10,000= ₹20.50

(NAV would stand reduced to the extent of dividend payout, being (10,000x10x10%) = ₹10,000)

Since dividend was reinvested by Mr. X, additional units acquired = $\frac{₹10,000}{₹\ 20.50}$ = 487.80 units

Therefore, units as on 31.03.2008 = 10, 000+ 487.80 = 10,487.80

Alternately, units as on 31.03.2008 = (2,15,000/20.50) = 10,487.80]

Dividend as on 31.03.2009 = 10,487.80 x 10 x 0.2 = ₹20,975.60

Let X be the NAV on 31.03.2009, then number of new units reinvested will be ₹ 20,975.60/X. Accordingly 11296.11 units shall consist of reinvested units and 10487.80 (as on 31.03.2008). Thus, by way of equation it can be shown as follows:

$$11296.11 = \frac{20975.60}{X} + 10487.80$$

Therefore, NAV as on 31.03.2009 = 20,975.60/(11,296.11- 10,487.80)

= ₹25.95

NAV as on 31.03.2010 = ₹ 1,00,000 (1+0.7352x33/12)/11296.11

= ₹ 26.75

Practical Q13

13. Mr. X on 1.7.2012, during the initial public offer of a Mutual Fund (MF) invested ₹ 1,00,000 at Face Value of ₹ 10. On 31.3.2013, the MF declared a dividend of 10% when Mr. X calculated that his holding period return was 115%. On 31.3.2014, MF again declared a dividend of 20%. On 31.3.2015, Mr. X redeemed all his investment which had accumulated to 11,296.11 units when his holding period return was 202.17%.

Calculate the NAVs as on 31.03.2013, 31.03.2014 and 31.03.2015.

Solution — ICAI

13. Yield for 9 months = 115%

Market value of Investments as on 31.03.2013 = 1,00,000 + (1,00,000 x 115%)

= ₹ 2,15,000/-

Therefore, NAV as on 31.03.2013 = (2,15,000 -10,000)/10,000 = ₹ 20.50

(NAV would stand reduced to the extent of dividend payout, being (₹100,000 x 10%)

= ₹ 10,000)

Since dividend was reinvested by Mr. X, additional units acquired

$$= \frac{₹\ 10,000}{₹\ 20.50} = 487.80 \text{ units}$$

Therefore, units as on 31.03.2013 = 10,000+ 487.80 = 10,487.80

[Alternately, units as on 31.03.2013 = (2,15,000/20.50) = 10,487.80]

Dividend as on 31.03.2014 = 10,487.80 x 10 x 0.2 = ₹ 20,975.60

Let X be the NAV on 31.03.2014, then number of new units reinvested will be ₹ 20,975.60/X. Accordingly 11296.11 units shall consist of reinvested units and 10487.80 (as on 31.03.2013). Thus, by way of equation it can be shown as follows:

$$11296.11 = \frac{20975.60}{X} + 10487.80$$

Therefore, NAV as on 31.03.2014 = 20,975.60/(11,296.11- 10,487.80)

= ₹ 25.95

NAV as on 31.03.2015 = ₹ 1,00,000 (1+2.0217)/11296.11

= ₹ 26.75

Practical Q14

14. A Mutual Fund having 300 units has shown its NAV of ₹ 8.75 and ₹ 9.45 at the beginning and at the end of the year respectively. The Mutual Fund has given two options:

(i) Pay ₹ 0.75 per unit as dividend and ₹ 0.60 per unit as a capital gain, or

(ii) These distributions are to be reinvested at an average NAV of ₹ 8.65 per unit.

What difference would it make in terms of return available and which option is preferable?

Solution — ICAI

14. **(i)** Returns for the year

(All changes on a Per -Unit Basis)

Change in Price:	₹ 9.45 – ₹8.75 = ₹ 0.70
Dividends received:	₹ 0.75
Capital gains distribution	₹ 0.60
Total reward	₹ 2.05
Holding period reward:	$\frac{₹2.05}{₹8.75} \times 100 = 23.43\%$

(ii) When all dividends and capital gains distributions are re-invested into additional units of the fund @ (₹ 8.65/unit)

= Dividend + Capital Gains per unit

= ₹ 0.75 + ₹ 0.60 = ₹ 1.35

Total received from 300 units = ₹1.35 x 300 = ₹405/-.

Additional Units Acquired

= ₹405/₹8.65 = 46.82 Units.

Total No.of Units_ = 300 units + 46.82 units = 346.82 units.

Value of 346.82 units held at the end of the year

= 346.82 units x ₹9.45 = ₹3277.45

Price Paid for 300 Units at the beginning of the year

= 300 units x ₹8.75 = ₹2,625.00

Holding Period Reward

₹ (3277.45 – 2625.00) = ₹652.45

Holding Period Reward $= \frac{₹\ 652.45}{₹\ 2625.00} \times 100 = 24.85\%$

Conclusion: Since the holding period reward is more in terms of percentage in option-two i.e., reinvestment of distributions at an average NAV of ₹8.65 per unit, this option is preferable.

Practical QIS

15. On 1-4-2012 ABC Mutual Fund issued 20 lakh units at ₹ 10 per unit. Relevant initial expenses involved were ₹ 12 lakhs. It invested the funds so raised in capital market instruments to build a portfolio of ₹ 185 lakhs. During the month of April 2012 it disposed off some of the instruments costing ₹ 60 lakhs for ₹ 63 lakhs and used the proceeds in purchasing securities for ₹ 56 lakhs. Fund management expenses for the month of April 2012 was ₹ 8 lakhs of which 10% was in arrears. In April 2012 the fund earned dividends amounting to ₹ 2 lakhs and it distributed 80% of the realized earnings. On 30-4-2012 the market value of the portfolio was ₹ 198 lakhs.

Mr. Akash, an investor, subscribed to 100 units on 1-4-2012 and disposed off the same at closing NAV on 30-4-2012. What was his annual rate of earning?

Solution — ICAI

15.

	Amount in ₹ lakhs	Amount in ₹ lakhs	Amount in ₹ lakhs
Opening Bank (200 - 185 -12)	3.00		
Add: Proceeds from sale of securities	63.00		
Add: Dividend received	2.00	68.00	
Deduct:			
Cost of securities purchased	56.00		
Fund management expenses paid (90% of 8)	7.20		
Capital gains distributed = 80% of (63 – 60)	2.40		
Dividend distributed =80% of 2.00	1.60	67.20	
Closing Bank			0.80
Closing market value of portfolio			198.00
			198.80
Less: Arrears of expenses			0.80
Closing Net Assets			198.00
Number of units (Lakhs)			20
Closing NAV per unit (198.00/20)			9.90

Rate of Earning (Per Unit)

	Amount
Income received (₹ 2.40 + ₹ 1.60)/20	₹ 0.20
Loss: Loss on disposal (₹ 200 - ₹ 198)/20	₹ 0.10
Net earning	₹ 0.10
Initial investment	₹ 10.00
Rate of earning (monthly)	1%
Rate of earning (Annual)	12%

Practical Q16

16. Sun Moon Mutual Fund (Approved Mutual Fund) sponsored open-ended equity-oriented scheme "Chanakya Opportunity Fund". There were three plans viz. 'A' – Dividend Re-investment Plan, 'B' – Bonus Plan & 'C' – Growth Plan.

At the time of Initial Public Offer on 1.4.1999, Mr. Anand, Mr. Bacchan & Mrs. Charu, three investors invested ₹ 1,00,000 each & chose 'B', 'C' & 'A' Plan respectively.

The History of the Fund is as follows:

Date	Dividend %	Bonus Ratio	Net Asset Value per Unit (F.V. ₹ 10)		
			Plan A	Plan B	Plan C

28.07.2003	20		30.70	31.40	33.42
31.03.2004	70	5 : 4	58.42	31.05	70.05
31.10.2007	40		42.18	25.02	56.15
15.03.2008	25		46.45	29.10	64.28
31.03.2008		1 : 3	42.18	20.05	60.12
24.03.2009	40	1 : 4	48.10	19.95	72.40
31.07.2009			53.75	22.98	82.07

On 31st July all three investors redeemed all the balance units.

Calculate annual rate of return to each of the investors.

Consider:

1. Long-term Capital Gain is exempt from Income tax.
2. Short-term Capital Gain is subject to 10% Income tax.
3. Security Transaction Tax 0.2 per cent only on sale/redemption of units.
4. Ignore Education Cess.

Solution — ICAI

16. **Mrs. Charu Plan A Dividend Reinvestment** *(Amount in ₹)*

Date	Investment (₹)	Dividend payout (%)	Dividend Re-invested (Closing Units X Face value of '10 X Dividend Payout %)	NAV (₹)	Units	Closing Unit Balance ∑ Units
01.04.1999	1,00,000.00			10.00	10,000.00	10,000.00
28.07.2003		20	20,000.00	30.70	651.47	10,651.47
31.03.2004		70	74,560.29	58.42	1,276.28	11,927.75
30.10.2007		40	47,711.00	42.18	1,131.13	13,058.88
15.03.2008		25	32,647.20	46.45	702.85	13,761.73
24.03.2009		40	55,046.92	48.10	1,144.43	14,906.16
Redemption value 14,906.16 × 53.75						8,01,206.10
Less: Security Transaction Tax (STT) is 0.2%						1,602.41
Net amount received						7,99,603.69
Less: Short term capital gain tax @ 10% on 1,144.43 (53.64* – 48.10≈) = 6,340						634
Net of tax						7,98,969.69
Less: Investment						1,00,000.00
						6,98,969.69

*(53.75 – STT @ 0.2%)

≈ This value can also be taken as zero

$$\text{Annual average return (\%)} = \frac{6,98,969.69}{1,00,000} \times \frac{12}{124} \times 100 = 67.64\ \%$$

Mr. Anand Plan B – Bonus

(Amount in ₹)

Date	Units	Bonus units	Total Balance	NAV per unit
01.04.1999	10,000		10,000	10
31.03.2004		12,500	22,500	31.05
31.03.2008		7,500	30,000	20.05
24.03.2009		7,500	37,500	19.95

Redemption value 37,500 × 22.98	8,61,750.00
Less: Security Transaction Tax (STT) is 0.2%	1,723.50
Net amount received	8,60,026.50
Less: Short term capital gain tax @ 10%	
7,500 × (22.93† – 19.95≈) = 22,350	2,235.00
Net of tax	8,57,791.50
Less: Investment	1,00,000.00
Net gain	7,57,791.50

†(22.98 – STT @ 0.2%)

≈ This value can also be taken as zero

$$\text{Annual average return (\%)} = \frac{7,57,791.50}{1,00,000} \times \frac{12}{124} \times 100 = 73.33\ \%$$

Mr. Bacchan Plan C – Growth

Particulars	*(Amount in ₹)*
Redemption value 10,000 × 82.07	8,20,700.00
Less: Security Transaction Tax (S.T.T) is .2%	1,641.40
Net amount received	8,19,058.60
Less: Short term capital gain tax @ 10%	0.00
Net of tax	8,19,058.60
Less: Investment	1,00,000.00
Net gain	7,19,058.60

$$\text{Annual average return (\%)} = \frac{7,19,058}{1,00,000} \times \frac{12}{124} \times 100 = 69.59\%$$

Note: Alternatively, figure of * and † can be taken as without net of Tax because, as per Proviso 5 of Section 48 of IT Act, no deduction of STT shall be allowed in computation of Capital Gain.

Practical Q17

17. A mutual fund company introduces two schemes i.e. Dividend plan (Plan-D) and Bonus plan (Plan-B). The face value of the unit is ₹ 10. On 1-4-2005 Mr. K invested ₹ 2,00,000 each in Plan-D and Plan-B when the NAV was ₹ 38.20 and ₹ 35.60 respectively. Both the plans matured on 31-3-2010.

Particulars of dividend and bonus declared over the period are as follows:

Date	Dividend	Bonus Ratio	Net Asset Value (₹)	
	%		Plan D	Plan B
30-09-2005	10	---	39.10	35.60
30-06-2006	---	1:5	41.15	36.25
31-03-2007	15	---	44.20	33.10
15-09-2008	13	---	45.05	37.25
30-10-2008	---	1:8	42.70	38.30
27-03-2009	16	---	44.80	39.10
11-04-2009	---	1:10	40.25	38.90
31-03-2010	---	---	40.40	39.70

What is the effective yield per annum in respect of the above two plans?

Solution — ICAI

17. **Plan – D**

Unit acquired = $\frac{2,00,000}{38.20}$ = 5235.60

Date	Units held	Dividend		Reinvestment	New	Total
		%	Amount	Rate	Units	Units
01.04.2005						5235.60
30.09.2005	5235.60	10	5235.60	39.10	133.90	5369.50
31.03.2007	5369.50	15	8054.25	44.20	182.22	5551.72
15.09.2008	5551.72	13	7217.24	45.05	160.20	5711.92
27.03.2009	5711.92	16	9139.07	44.80	204	5915.92
31.03.2010	Maturity Value		(₹ 40.40 X 5915.92)			₹ 2,39,003.17
	Less: Cost of Acquisition					₹ 2,00,000.00
	Total Gain					₹ 39,003.17

$$\therefore \text{Effective Yield} = \frac{₹39,003.17}{₹2,00,000} \times \frac{1}{5} \times 100 = 3.90\%$$

Alternatively, it can be computed by using the IRR method as follows:

NPV at 4% = - ₹ 2,00,000 + ₹ 1,96,443 = - ₹ 3,557

NPV at 2% = - ₹ 2,00,000 + ₹ 2,16,473 = ₹ 16,473

$$\text{IRR} = \text{LR} + \frac{\text{NPV at LR}}{\text{NPV at LR - NPV at HR}} (\text{HR - LR}) = 2\% + \frac{16473}{16473 - (-3557)} (4\% - 2\%) = 3.645\%$$

Plan – B

Date	Particulars	Calculation Working	No. of Units	NAV (₹)
1.4.05	Investment	₹2,00,000/35.60=	5617.98	35.60
30.6.06	Bonus	5617.98/5 =	1123.60	36.25
			6741.58	
30.10.08	"	6741.58/8 =	842.70	38.30
			7584.28	
11.4.09	"	7584.28/10 =	758.43	38.90
			8342.71	
31.3.10	Maturity Value	8342.71 x ₹ 39.70=		3,31,205.59
	Less: Investment			2,00,000.00
	Gain			1,31,205.59

$$\therefore \text{Effective Yield} = \frac{1,31,205.59}{2,00,000} \times \frac{1}{5} \times 100 = 13.12\%$$

Alternatively, it can be computed by using the IRR method as follows:

NPV at 13% = - ₹ 2,00,000 + ₹ 1,79,765 = - ₹ 20,235

NPV at 8% = - ₹ 2,00,000 + ₹ 2,25,413 = ₹ 25,413

$$\text{IRR} = \text{LR} + \frac{\text{NPV at LR}}{\text{NPV at LR - NPV at HR}} (\text{HR - LR}) = 8\% + \frac{25413}{25413 - (-20235)} (13\% - 8\%) = 10.78\%$$

Practical Q18

18. A mutual fund made an issue of 10,00,000 units of ₹ 10 each on January 01, 2008. No entry load was charged. It made the following investments:

Particulars	₹
50,000 Equity shares of ₹ 100 each @ ₹ 160	80,00,000
7% Government Securities	8,00,000
9% Debentures (Unlisted)	5,00,000
10% Debentures (Listed)	5,00,000
	98,00,000

During the year, dividends of ₹ 12,00,000 were received on equity shares. Interest on all types of debt securities was received as and when due. At the end of the year equity shares and 10% debentures are quoted at 175% and 90% respectively of face value. Other investments are valued at par.

Find out the Net Asset Value (NAV) per unit given that operating expenses paid during the year amounted to ₹ 5,00,000. Also find out the NAV, if the Mutual fund had distributed a dividend of ₹ 0.80 per unit during the year to the unit holders.

Solution — ICAI

18. In order to find out the NAV, the cash balance at the end of the year is calculated as follows-

Particulars	₹
Cash balance in the beginning	
(₹ 100 lakhs – ₹ 98 lakhs)	2,00,000
Dividend Received	12,00,000
Interest on 7% Govt. Securities	56,000
Interest on 9% Debentures	45,000
Interest on 10% Debentures	50,000
	15,51,000
(-) Operating expenses	5,00,000
Net cash balance at the end	10,51,000
Calculation of NAV	₹
Cash Balance	10,51,000
7% Govt. Securities (at par)	8,00,000
50,000 equity shares @ ₹ 175 each	87,50,000
9% Debentures (Unlisted) at cost	5,00,000
10% Debentures @90%	4,50,000
Total Assets	1,15,51000
No. of Units	10,00,000
NAV per Unit	₹ 11.55

Calculation of NAV, if dividend of ₹ 0.80 is paid –

Net Assets (₹ 1,15,51,000 – ₹ 8,00,000)	₹ 1,07,51,000
No. of Units	10,00,000
NAV per unit	₹ 10.75

Practical Q19

19. Based on the following information, determine the NAV of a regular income scheme on per unit basis:

Particulars	₹ Crores
Listed shares at Cost (ex-dividend)	20
Cash in hand	1.23
Bonds and debentures at cost	4.3
Of these, bonds not listed and quoted	1
Other fixed interest securities at cost	4.5
Dividend accrued	0.8
Amount payable on shares	6.32
Expenditure accrued	0.75
Number of units (₹ 10 face value)	20 lacs
Current realizable value of fixed income securities of face value of ₹ 100	106.50
The listed shares were purchased when Index was	1,000
Present index is	2,300
Value of listed bonds and debentures at NAV date	8

There has been a diminution of 20% in unlisted bonds and debentures. Other fixed interest securities are at cost.

Solution — ICAI

19.

Particulars	Adjusted Values ₹ crores
Equity Shares	46.00
Cash in hand	1.23
Bonds and debentures not listed	0.80
Bonds and debentures listed	8.00
Dividends accrued	0.80
Fixed income securities	4.50
Sub total assets (A)	**61.33**
***Less:* Liabilities**	
Amount payable on shares	6.32
Expenditure accrued	0.75
Sub total liabilities (B)	**7.07**
Net Assets Value (A) – (B)	**54.26**
No. of units	20,00,000
Net Assets Value per unit (₹ 54.26 crore / 20,00,000)	₹ 271.30

Practical Q20

20. On 1st April, an open ended scheme of mutual fund had 300 lakh units outstanding with Net Assets Value (NAV) of ₹ 18.75. At the end of April, it issued 6 lakh units at opening NAV plus 2% load, adjusted for dividend equalization. At the end of May, 3 Lakh units were repurchased at opening NAV less 2% exit load adjusted for dividend equalization. At the end of June, 70% of its available income was distributed.

In respect of April-June quarter, the following additional information are available:

	₹ in lakh
Portfolio value appreciation	425.47
Income of April	22.950
Income for May	34.425
Income for June	45.450

You are required to calculate:

(i) Income available for distribution;

(ii) Issue price at the end of April;

(iii) repurchase price at the end of May; and

(iv) net asset value (NAV) as on 30th June.

Solution — ICAI

20. Calculation of Income available for Distribution

	Units (Lakh)	Per Unit (₹)	Total (₹ In lakh)
Income from April	300	0.0765	22.9500
Add: Dividend equalization collected on issue	6	0.0765	0.4590
	306	0.0765	23.4090
Add: Income from May		0.1125	34.4250
	306	0.1890	57.8340
Less: Dividend equalization paid on repurchase	3	0.1890	(0.5670)
	303	0.1890	57.2670
Add: Income from June		0.1500	45.4500
	303	0.3390	102.7170
Less: Dividend Paid		0.2373	(71.9019)
	303	0.1017	30.8151

Calculation of Issue Price at the end of April

	₹
Opening NAV	18.750
Add: Entry Load 2% of ₹ 18.750	0.375
	19.125
Add: Dividend Equalization paid on Issue Price	0.0765
	19.2015

Calculation of Repurchase Price at the end of May

	₹
Opening NAV	18.750
Less: Exit Load 2% of ₹ 18.750	(0.375)
	18.375
Add: Dividend Equalization paid on Issue Price	0.1890
	18.564

Closing NAV

		₹ (Lakh)
Opening Net Asset Value (₹ 18.75 × 300)		5625.0000
Portfolio Value Appreciation		425.4700
Issue of Fresh Units (6 × 19.2015)		115.2090
Income Received (22.950 + 34.425 + 45.450)		102.8250
		6268.504
Less: Units repurchased (3 × 18.564)	-55.692	
Income Distributed	-71.9019	(-127.5939)
Closing Net Asset Value		6140.9101
Closing Units (300 + 6 – 3) lakh		303 lakh
∴ Closing NAV as on 30th June		₹ 20.2670

Practical Q21

21. There are two Mutual Funds viz. D Mutual Fund Ltd. and K Mutual Fund Ltd. Each having close ended equity schemes.

NAV as on 31-12-2014 of equity schemes of D Mutual Fund Ltd. is ₹ 70.71 (consisting 99% equity and remaining cash balance) and that of K Mutual Fund Ltd. is ₹ 62.50 (consisting 96% equity and balance in cash).

Following is the other information:

Particular	Equity Schemes	
	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Sharpe Ratio	2	3.3
Treynor Ratio	15	15
Standard deviation	11.25	5

There is no change in portfolios during the next month and annual average cost is ₹ 3 per unit for the schemes of both the Mutual Funds.

If Share Market goes down by 5% within a month, calculate expected NAV after a month for the schemes of both the Mutual Funds.

For calculation, consider 12 months in a year and ignore number of days for particular month.

Solution — ICAI

21. Working Notes:

(i) Decomposition of Funds in Equity and Cash Components

	D Mutual Fund Ltd.	K Mutual Fund Ltd.
NAV on 31.12.14	₹ 70.71	₹ 62.50
% of Equity	99%	96%
Equity element in NAV	₹ 70	₹ 60
Cash element in NAV	₹ 0.71	₹ 2.50

(ii) Calculation of Beta

(a) D Mutual Fund Ltd.

$$\text{Sharpe Ratio} = 2 = \frac{E(R) - R_f}{\sigma_D} = \frac{E(R) - R_f}{11.25}$$

$$E(R) - R_f = 22.50$$

$$\text{Treynor Ratio} = 15 = \frac{E(R) - R_f}{\beta_D} = \frac{22.50}{\beta_D}$$

$$\beta_D = 22.50/15 = 1.50$$

(b) K Mutual Fund Ltd.

$$\text{Sharpe Ratio} = 3.3 = \frac{E(R) - R_f}{\sigma_K} = \frac{E(R) - R_f}{5}$$

$$E(R) - R_f = 16.50$$

$$\text{Treynor Ratio} = 15 = \frac{E(R) - R_f}{\beta_K} = \frac{16.50}{\beta_K}$$

$$\beta_K = 16.50/15 = 1.10$$

(iii) Decrease in the Value of Equity

	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Market goes down by	5.00%	5.00%
Beta	1.50	1.10
Equity component goes down	7.50%	5.50%

(iv) Balance of Cash after 1 month

	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Cash in Hand on 31.12.14	₹ 0.71	₹ 2.50
Less: Exp. Per month	₹ 0.25	₹ 0.25
Balance after 1 month	₹ 0.46	₹ 2.25

NAV after 1 month

	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Value of Equity after 1 month		
70 x (1 - 0.075)	₹ 64.75	-
60 x (1 - 0.055)	-	₹ 56.70
Cash Balance	0.46	2.25
	65.21	58.95

Practical Q22

22. ANP Plan, a hedge fund currently has assets of ₹ 20 crore. CA. X, the manager of fund charges fee of 0.10% of portfolio asset. In addition to it he charges incentive fee of 2%. The incentive will be linked to gross return each year in excess of the portfolio maximum value since the inception of fund. The maximum value the fund achieved so far since inception of fund about one and half year ago was ₹ 21 crores.

You are required to compute the fee payable to CA. X, if return on the fund this year turns out to be

(a) 29%, (b) 4.5%, (c) -1.8%

Solution — ICAI

22. (a) If return is 29%

	₹
Fixed fee (A) 0.10% of ₹ 20 crore	2,00,000
New Fund Value (1.29 x ₹ 20 crore)	25.80 crore
Excess Value of best achieved (25.8 crore – 21.0 crore)	4.80 crore
Incentive Fee (2% of 4.80 crores) (B)	9,60,000
Total Fee (A)+(B)	11,60,000

(b) If return is 4.5%

	₹
Fixed (A) 0.10% of ₹ 20 crore	2,00,000
New Fund Value (1.045 x ₹ 20 crore)	20.90 crore
Excess Value of best achieved (20.90 crore –21.00 crore)	(₹ 0.10 crore)
Incentive Fee (as does not exceed best achieved) (B)	Nil
Total Fee (A)+(B)	2,00,000

(c) If return is (-1.8%)

No incentive only fixed fee of ₹ 2,00,000 will be paid

Practical Q23

23. Ms. Sunidhi is working with an MNC at Mumbai. She is well versant with the portfolio management techniques and wants to test one of the techniques on an equity fund she has constructed and compare the gains and losses from the technique with those from a passive buy and hold strategy. The fund consists of equities only and the ending NAVs of the fund he constructed for the last 10 months are given below:

Month Ending	NAV (₹/unit)	Month Ending	NAV (₹/unit)
December 2008	40.00	May 2009	37.00
January 2009	25.00	June 2009	42.00
February 2009	36.00	July 2009	43.00
March 2009	32.00	August 2009	50.00
April 2009	38.00	September 2009	52.00

Assume Sunidhi had invested a notional amount of ₹ 2 lakhs equally in the equity fund and a conservative portfolio (of bonds) in the beginning of December 2008 and the total portfolio was being rebalanced each time the NAV of the fund increased or decreased by 15%.

You are required to determine the value of the portfolio for each level of NAV following the Constant Ratio Plan.

Solution — ICAI

23. Constant Ratio Plan:

Stock Portfolio NAV (₹)	Value of Conservative Portfolio (₹)	Value of aggressive Portfolio (₹)	Total value of Constant Ratio Plan (₹)	Revaluation Action	Total No. of units in aggressive portfolio
40.00	1,00,000	1,00,000	2,00,000	-	2500
25.00	1,00,000	62,500	1,62,500	-	2500
	81,250	81,250	1,62,500	Buy 750 units	3250
36.00	81,250	1,17,000	1,98,250	-	3250
	99,125	99,125	1,98,250	Sell 496.53 units	2753.47
32.00	99,125	88,111.04	1,87,236.04	-	2753.47
38.00	99,125	1,04,631.86	2,03,756.86	-	2753.47
	1,01,878.43	1,01,878.43	2,03,756.86	Sell 72.46 units	2681.01
37.00	1,01,878.43	99,197.37	2,01,075.80	-	2681.01
42.00	1,01,878.43	1,12,602.42	2,14,480.85	-	2681.01
43.00	1,01,878.43	1,15,283.43	2,17,161.86	-	2681.01
50.00	1,01,878.43	1,34,050.50	2,35,928.93	-	2681.01
	1,17,964.47	1,17,964.46	2,35,928.93	Sell 321.72 units	2359.29
52.00	1,17,964.47	1,22,683.08	2,40,647.55	-	2359.29

Hence, the ending value of the mechanical strategy is ₹ 2,40,647.55 and buy & hold strategy is ₹ 2,60,000.

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